Stillwaters AI

GENERATIVE AI FRONTIERS

BY STILLWATERS RESEARCH

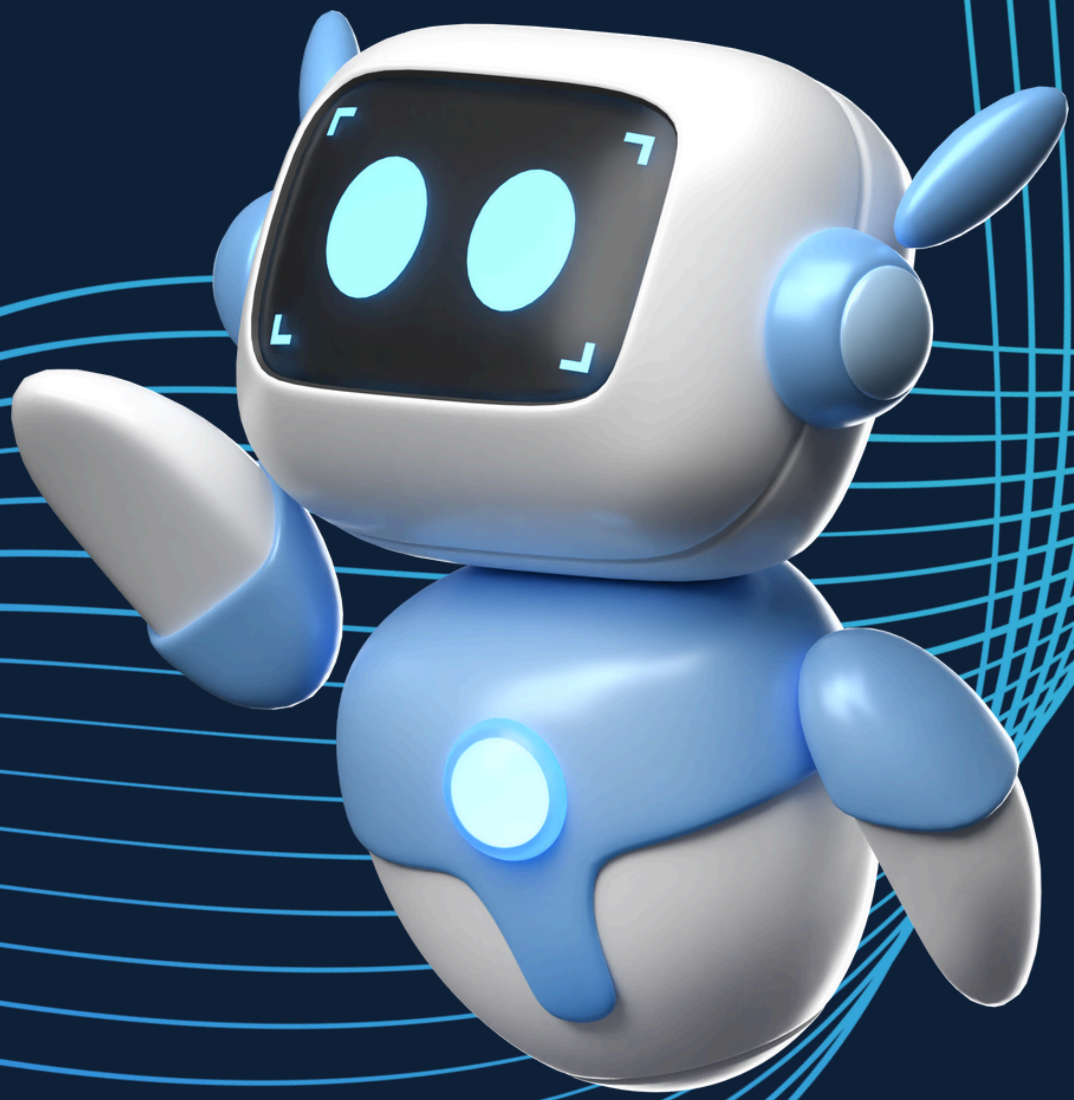


TABLE OF CONTENT

1. Introduction
2. What makes us different
3. Stillwaters ASR
4. Stillwaters Revolutionary Command Generation
   - SQL and MongoQL command generation
   - Multilingual Medical AI
5. Stillwaters Contextual Search for healthcare
6. Stillwaters Tracerunner
7. Stillwaters Generative AI Seminar Agenda

GENERATIVE AI FRONTIERS

Introduction:

At Stillwaters Research, we're leading the charge in a new era of artificial intelligence. Our mission is to redefine the boundaries of AI by focusing on smaller, highly specialized models that excel in specific tasks, outperforming the cumbersome, costly, and less secure Large Language Models (LLMs). While we continue to use LLMs where necessary, we are at the forefront of pioneering compact, powerful models—carving out a niche that sets us apart in the GenAI landscape.

WHAT MAKES US DIFFERENT

Stillwaters ASR

STT (Speech-to-Text) and TTS (Speech-to-Text) Excellence: With a focus on Indic languages, Stillwaters Research is at the pinnacle of accuracy for languages including English, Hindi, Telugu, Tamil, and Kannada. Our models deliver this precision with an impressively small GPU RAM footprint of just 1.5 GB.

STILLWATERS REVOLUTIONARY COMMAND GENERATION:

Our journey through Generative AI (GenAI) and traditional AI is nothing short of groundbreaking. We've achieved industry-leading accuracy with text-to-SQL conversions, boasting 99% execution accuracy and 93% contextual accuracy for both MongoQL and SQL. And we achieve this with an highly efficient 12-15 GB GPU RAM footprint—allowing our system to generate SQL and MongoQL with identical precision. To top it all it is **"zero shot learning".**

SQL and MongoQL command generation

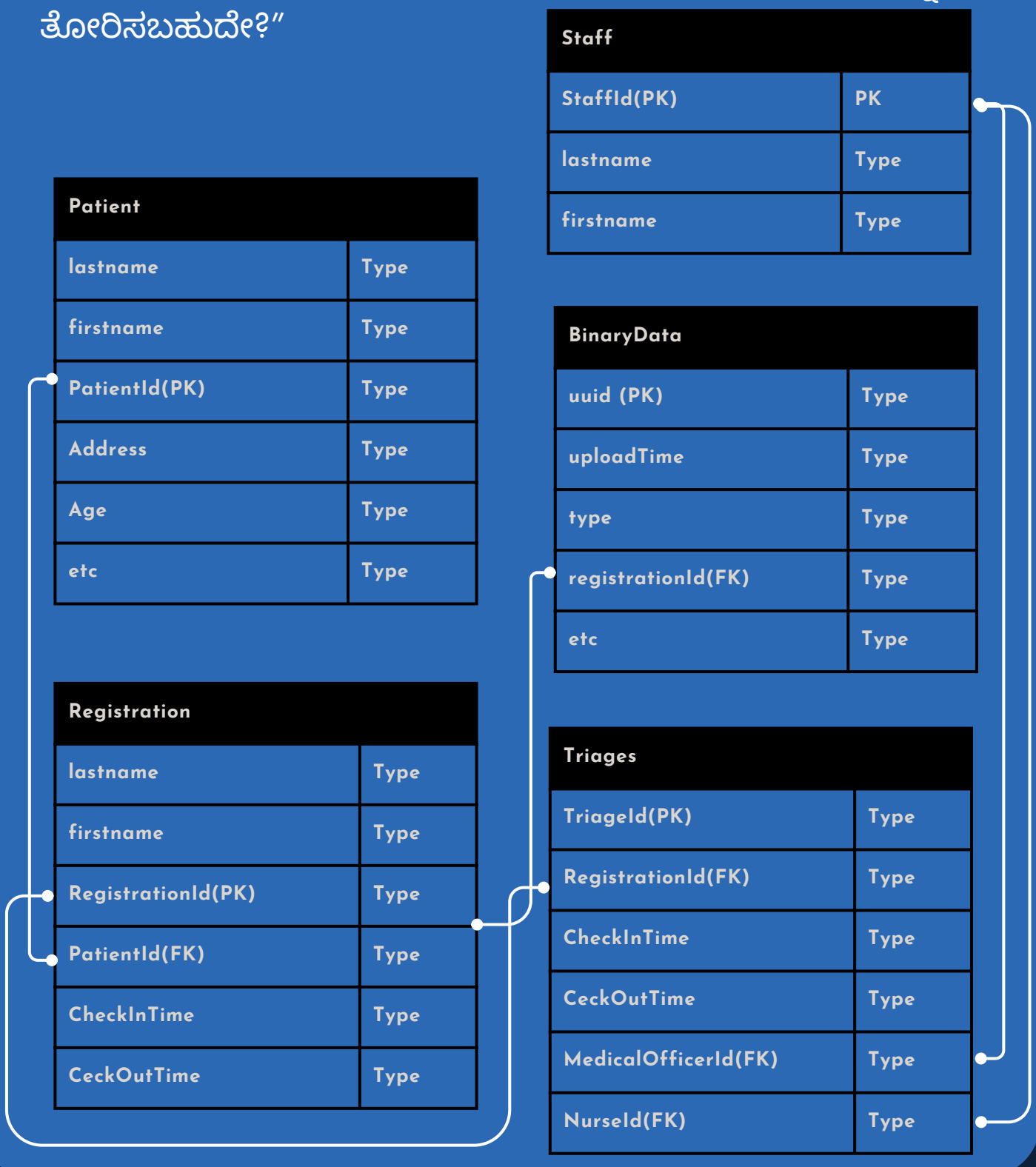
Consider below a simplified SQL structure and then follow along with the examples. **The queries may be expressed in voice and text and using all the above supported lanuages.** The language used in the **example below is very formal**, but **in practice** the language used **may be very very informal to the point of being slangy**, and a **single sentence may have multiple language**s as usually speak (Hinglish, Kanglish etc.).

Queries:

Eng: "Could you show me the nurse assigned to patient Keshav."

Hindi: “क्या आप मुझे मरीज केशव के लिए नियुक्त की गई नर्स को दिखा सकते हैं?”

Kannada: ”ನಾನು ಕೇಶವ್ ರೋಗಿಗೆ ನೇಮಕಗೊಂಡಿರುವ ನರ್ಸ್ ಅನ್ನು ತೋರಿಸಬಹುದೇ?”



SQL:

```
SELECT
 staff.firstName,
 staff.lastName

FROM
 triages AS triages

JOIN
 staff AS staff
 ON triages.medicalOfficerId = staff.guid

JOIN
 registrations AS registrations
 ON triages.registrationId =
registrations.registrationId

JOIN
 patients AS patients
 ON registrations.patientMedicalRecordNumber =
patients.medicalRecordNumber

WHERE
 patients.firstName LIKE 'keshav%'
```

MongoQL:

```
db.triages.aggregate([
  {
    "$lookup": {
      "from": "staff",
      "let": { "medicalofficerid":
"$medicalOfficerId" },
      "pipeline": [
        {
          "$match": {
            "$expr": { "$eq": [
"$$medicalofficerid", "$guid" ] }
          }
        }
      ],
      "as": "staff"
    }
  },
  {
    "$unwind": {
      "path": "$staff",
"preserveNullAndEmptyArrays": false
    }
  },
  {
    "$lookup": {
      "from": "registrations",
      "let": { "registrationid": "$registrationId" },
      "pipeline": [
        {
          "$match": {
            "$expr": { "$eq": [ "$$registrationid",
"$registrationId" ] }
```

```
        }
    },
    {
        "$unwind": {
            "path": "$registrations",
"preserveNullAndEmptyArrays": false
        }
    },
    {
        "$lookup": {
            "from": "patients",
            "let": {
"registrations_patientmedicalrecordnumber":
"$registrations.patientMedicalRecordNumber" },
            "pipeline": [
                {
                    "$match": {
                        "$and": [
                            {
"$expr": { "$eq": [
"$$registrations_patientmedicalrecordnumber",
"$medicalRecordNumber" ] }
                            },
                            {
"firstName": { "$regex": "^keshav.*$", "$options": "i"
}
                            }
                        ]
                    }
                }
            ],
```

```
            "as": "patients"
        }
    },
    {
        "$unwind": {
            "path": "$patients",
"preserveNullAndEmptyArrays": false
        }
    },
    {
        "$skip": 0
    },
    {
        "$limit": 5
    },
    {
        "$project": {
            "_id": 0,
            "staff.firstName": 1,
            "staff.lastName": 1,
            "patients.firstName": 1,
            "patients.lastName": 1
        }
    }
])
```

MULTILINGUAL MEDICAL AI:

As part of our upcoming Medical AI product launch in India, we're breaking new ground by supporting NLP queries in multiple languages—**Hindi, Telugu, Tamil, and Kannada**—with unparalleled accuracy. But we don't stop there. Our capabilities extend to **generic command recognition**, enabling intuitive interactions, like "play this song on spotify" to "profile <path to docker-compose or kubernetes configfile> application" making technology more accessible and user-friendly.

A Commitment to Excellence

At Stillwaters research, we are not just pushing boundaries—we are transforming them. Our dedication to excellence is turning possibilities into realities, proving that AI can be powerful, precise, and efficient without the need for vast resources. Join us in shaping the future of AI, where innovation meets precision in the most efficient space possible.

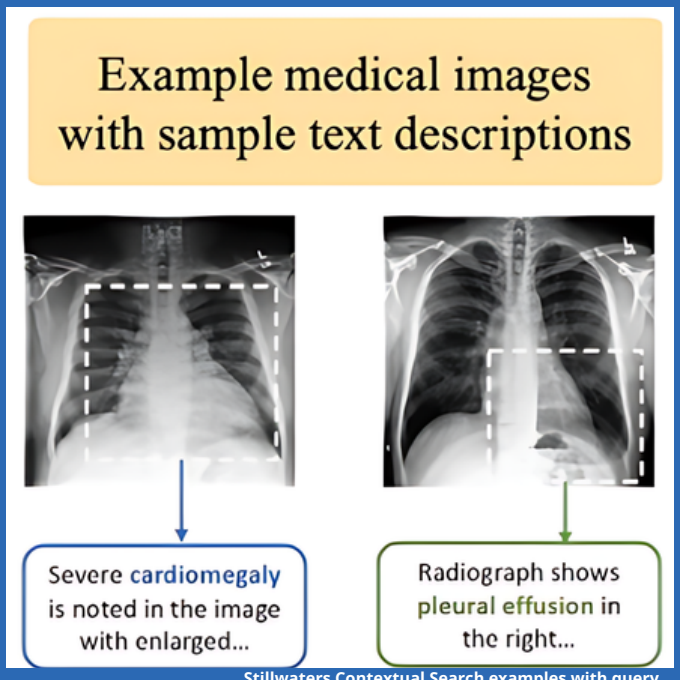
STILLWATERS CONTEXTUAL SEARCH FOR HEALTHCARE

We extend our contextual search to our medical AI product. Where a search for "PA (posteroanterior)" view of the hand or "pinch view" or "pinch grip view" X-ray fetches the below picture.



pinch view

A few more examples



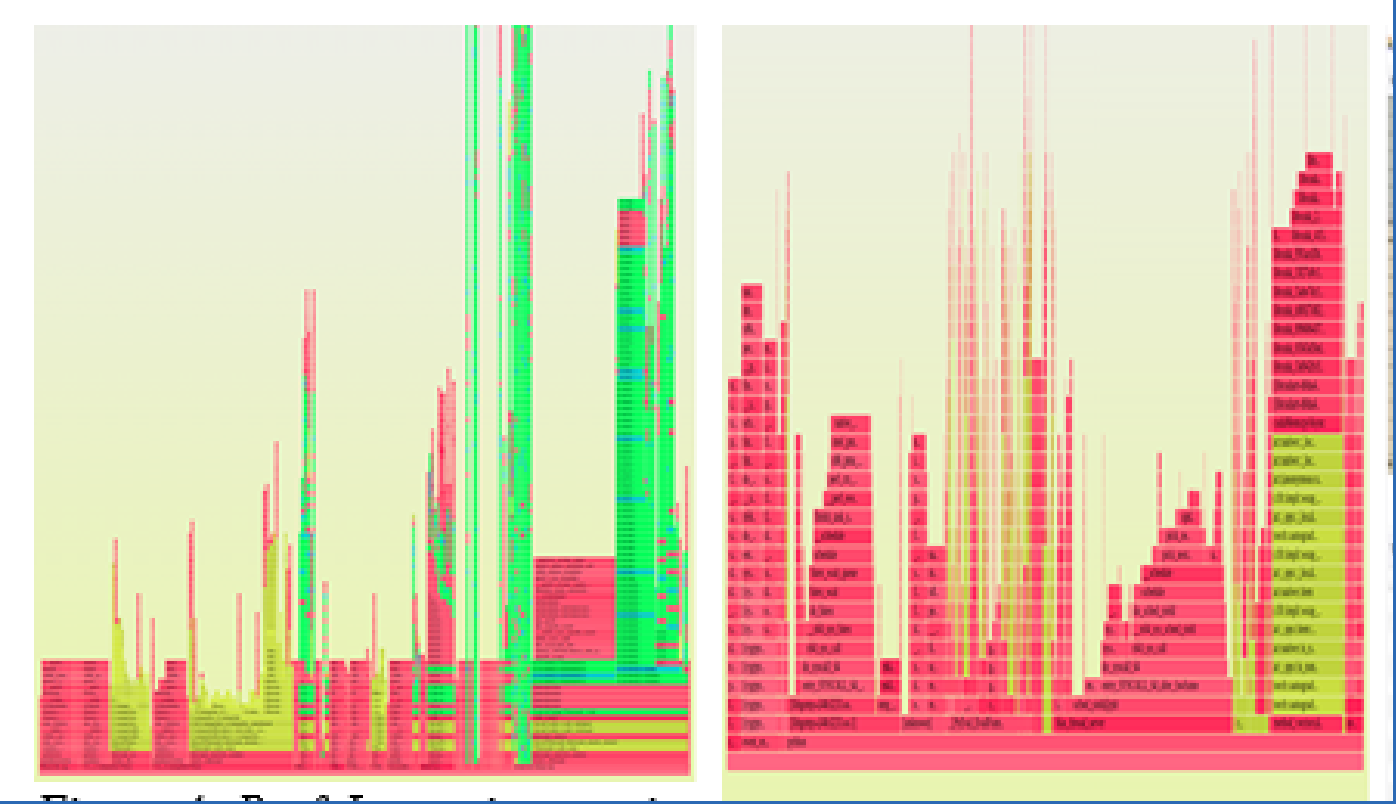
Stillwaters Contextual Search examples with query

STILLWATERS TRACERUNNER (STILLWATERS INTEGRATED PROFILING SERVICES)

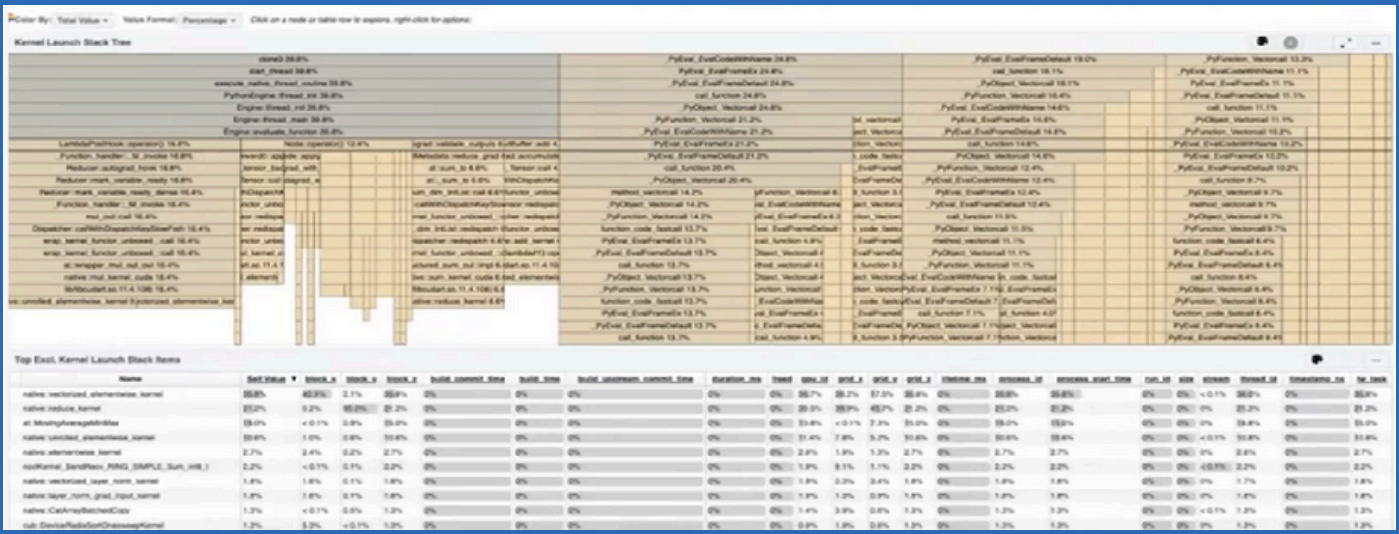
At Stillwaters.ai, we understand the critical need for an integrated, continuous profiling tool that includes GPU workloads. Existing tools, like Gupti or NVIDIA's CUDA profiling tool, have their limitations—requiring code instrumentation, running inside your process, and often lacking CPU context.

Enter **StillwatersTracerunner**. This tool seamlessly launches a sidecar from a Docker Compose or Kubernetes config, allowing continuous profiling of Java, Python, C++, and CUDA processes without impacting performance or reliability. It's **AI framework(tensorflow/pytorch/jax) agnostic,** making it the perfect solution for any platform. We use **lbr(intels lastBranchRecord), dwarf or framepointers for stack unwinding** wherever available.

The **StillwatersTracerunner is AI framework** agnostic.
Samples are shown below.



Perf-Java-microservice-profile **CUDA-profile**



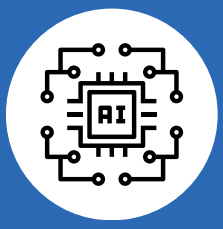
CUDA-Kernel

STILLWATERS GENERATIVE AI SEMINAR (90 MINUTES)

But how do you know the above claims are correct and we are not cherry picking. This is where you fall into our trap. Call us for a demo/seminar and see us live.

- Continous Profiling microservices environment including CUDA
- Transformer Architecture (15 minutes)
  - Overview of transformer models
  - Key components: encoder, decoder, attention mechanism
  - Applications in NLP, including text-to-SQL and text-to-MongoQL
- Incorporating Graph Inputs into Transformer Models
  - Understanding graph structures and their relevance in AI
  - Techniques for encoding graph data into transformers
  - Examples and use cases
- Leveraging Table Knowledge in Transformer Models
  - Importance of table knowledge in AI applications
  - Methods for integrating table data with transformers
  - Case studies and practical applications
- We intend to demonstrate and share our GenAI differentiators
  - Text to SQL/MongoQL/Command
  - Product Demo

WE CAN HELP YOU BUILD ARTIFICIAL INTELLIGENCE BASED SYSTEMS.



We can build it for you



We can help you build it with your team



We can train your organization in technologies like HPC, Deep Learning, NLP

For any enquiries or to book a free seminar on any of our modules, please get in touch:

relations@stillwaters.ai